The following Tamil puzzle is taken from the famous Mathematics problems collection book in Tamil called “Kanakkathikaram” in which is a good example of conversion of units of distance :

முப்பத்தி ரண்டு முழம்உளமுட் பனையை
தப்பாமல் ஒந்தி தவழ்ந்தேறிச் – செப்பமுடச்
சாணேறி நான்குவிரற்கிழியும் என்பரே
நாணா தொரு நாள் நகர்ந்து

Meaning of this puzzle:

On a palm tree of height 32 cubit , a chameleon tried to reach the top of the tree. If it climbed one hand span on it but slipped four fingers on one day, in how many days will it reach the top of the tree?

Solution:

1 hand span = 12 fingers;
1 cubit = 2 hand spans = 24 fingers.
Height of the palm tree = 32 cubit = 32 x 24 fingers = 768 fingers.
Distance climbed in 1 day = 1 hand span = 12 fingers.
Distance slipped in 1 day = 4 fingers.
Actual distance climbed in 1 day = 12 – 4 = 8 fingers.
Number of days required to climb the top of the tree = 768/ 8 = 96 days

2.4 Fundamental Operations on Quantities with Different Units

We can do the basic operations on the metric units as we do the decimal operations. Note that, measurements with the same unit can be added/ subtracted, but unlike units of measurements should be converted into like units and then they can be added / subtracted.

Example 8: Saritha bought 6 m and 40 cm of cloth for herself and 3 m and 80 cm of cloth for her sister. What was the total length of the cloth bought by her?

Solution:

	m	*cm*
	1	
Length of cloth bought for Saritha	6	40
Length of cloth bought for Saritha’s sister	3	80
Total length of the cloth	10m	20cm

Example 9: Pradeep travels 4 km and 350 m to reach to the market, while Kandan travels 6 km and 200 m to reach to the same market from their houses. How much distance does Kandan travel more than Pradeep?

Solution:

	km	*m*
	1000 m →	
Distance travelled by Kandan	~~6~~ 5	~~200~~ 1200
Distance travelled by Pradeep	4	350
Difference in distance of their travel	1*km*	850*m*

Kandan travelled 1 km and 850 m more than Pradeep.

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